D06 · Logistics Technology & Innovation

01 · Logistics technology stack: TMS, WMS & visibility platforms

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1 WHAT IT IS

The logistics technology stack is the set of execution systems — WMS for the warehouse, TMS for transport, visibility platforms for in-transit events — and the integrations that connect them to each other and to the ERP.

2 WHY IT MATTERS

Each system creates value only when data flows cleanly between them: **a WMS without ERP integration creates two inventories; a TMS without WMS confirmation plans loads that do not exist**. The choice between ERP-native modules and best-of-breed products is a trade-off between integration simplicity and functional depth, and the integration cost is the item most often underestimated.

3 HOW IT IS DONE

1. **Map processes before products.** Define which decisions each system owns and which data each one masters.
2. **Decide native vs. best-of-breed on complexity.** Simple, ERP-centric operations favor native; complex, specialized ones may justify best-of-breed.
3. **Sequence implementations.** Usually WMS first (inventory truth), then TMS, then the visibility layer.
4. **Budget integration and change management seriously.** Interfaces, testing and training are a large share of total cost and of the result.

IN PRACTICE

Prioritize clean integration over feature richness

A simpler system that is fully integrated almost always outperforms a feature-rich system that requires constant manual intervention.

Sequence implementations deliberately

WMS -> TMS -> Visibility is the lowest-risk order in the majority of environments.

Budget seriously for change management and process redesign

Technology that is used as a recording system instead of a decision and optimization system delivers a fraction of its potential.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Distribution company with ERP + third-party WMS but no TMS or visibility

Gap	Impact	Solution
No TMS	Manual booking, unaudited invoices, low fill rates	SAP TM (native) or best-of-breed TMS
No multi-carrier visibility	Reactive exception handling, customer complaints	project44 / equivalent visibility layer
Manual ERP–WMS sync	Daily reconciliation effort, inventory discrepancies	Real-time integration

Result: Business case showed 9-month payback from headcount release, freight recovery and reduced expedites. Implementation sequenced WMS integration first, then TMS, then visibility.

HOW IT IS MEASURED

Systems Integration Rate %

(Logistics transactions processed without manual re-entry / Total logistics transactions) × 100

Benchmark: >85% is a realistic target for a well-integrated stack

Technology ROI (3-year)

(Quantified benefits – total cost of ownership) / total cost of ownership

Benchmark: Target >200–300% over three years for major stack investments

WHAT YOU WOULD USE

Core Stack

SAP EWM + SAP TM — Strong choice when the company is already on S/4HANA and values native integration.

Manhattan / Blue Yonder / Körber (WMS) + specialized TMS + project44/FourKites — Common best-of-breed combination for complex operations.

4 SO WHAT

Buy integrations, not features. Assign data ownership, choose native or best-of-breed by complexity, sequence WMS -> TMS -> visibility, and fund the change management that makes systems change outcomes.

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